DREADNOUGHT EXPLORATION PTY LTD

CO-FUNDED DRILLING REPORT MANGAROON CARBONATITES

For the Period

1 January 2024 to 31 December 2024

Co-funded drilling report Mangaroon REE Project (R28)

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ATTACHMENTS SUBMITTED SEPARATELY

1. Bibliographic Data Sheet

Project Name:	Mangaroon Carbonatites
Combined Reporting Number:	C108/2022
Tenement Numbers:	E09/02448,E09/02450
Tenement Operator(s):	DREADNOUGHT EXPLORATION PTY LTD
Report Type:	Co-Funded Drilling
Report Title:	Co-funded drilling report Mangaroon REE Project (R28)
Report Period:	1 January 2024 to 31 December 2024
Author:	Nicholas CHAPMAN
Submitted By:	Claudia TOMKINS
Report Date:	27 November 2024

Map Sheets:	*1:250,000 Map Sheet*	*1:100,000 Map Sheet*
	SF50-14 (EDMUND)	2150 (EDMUND)
	SG50-02 (MOUNT PHILLIPS)	2149 (MOUNT PHILLIPS)

Target Commodity:	NIOBIUM, RARE EARTH ELEMENTS
Prospects Drilled:	C2, C3, C5, Stinger
PoW Number:	
Geophysical Survey Reg No:	
Assays:	

Abstract

Location: The GCC is part of the wider Mangaroon Project, a collection of 30 tenements spanning ~4800km^2. The Mangaroon Project is situated approximately 250 kilometers to the southeast of Exmouth, at the confluence of the Upper Gascoyne, Carnarvon, and Ashburton shires, within Western Australia's Gascoyne Region.

Access to the project area can be obtained through several routes: from Carnarvon or Gascoyne Junction via the Ullawarra/Gifford Creek Road, from Exmouth via the Towera Road branching off the Great Northern Highway, or from Meekatharra via Mount Augustus along national route 95. The terrain consists of gently undulating landscapes featuring incised drainage patterns and multiple creek crossings accessible by four-wheel drive vehicles.

The GCC, where RC drilling was completed, is located ~3km to the NW of, and parallel to the Cobra-Gifford Creek Road. The GCC extends 17km southeast of the entrance to the Yangibana Mine ('Borrow Pit'), ending just north of the Gifford Creek homestead.

Geology: The Mangaroon REE Project lies in the Gascoyne region of Western Australia between the Yilgarn Craton and the Pilbara Craton. The Gascoyne geological province consists of a basement of Neoarchean to Palaeoproterozoic granitic gneisses (the Glenburgh Terrane) overlain by various Proterozoic metasedimentary and granitic rocks (Johnson et. al 2011). The Glenburgh Terrane was accreted to the Pilbara Craton during the Ophthalmian Orogeny (c. 2215–2145 Ma). The combined Pilbara Craton and Glenburgh Terrane then collided with the Yilgarn craton during the Glenburgh Orogeny (c. 1950 Ma). Subsequent regional deformation occurred during the Capricorn Orogeny (c. 1820–1770 Ma) and Mangaroon Orogeny (c. 1680–1620 Ma) (Sheppard et. al 2010, Sheppard et. al 2016). The Gascoyne Province has recorded six tectono-thermal events between 2005–570 Ma and contains numerous crustal-scale faults and shear zones.

Carbonatite dykes, termed the Gifford Creek Carbonatite Complex (GCCC), were emplaced in the Gascoyne region c. 1370 Ma utilising the Lyons River Fault as a conduit through the crust (Slezak and Spandler 2020). The GCCC is a REE-prospective unit which contains the Yangibana Ironstones—REE mineralised carbonatite dykes emplaced northeast of the Lyons River Fault hosting a 29.93Mt @ 0.94% TREO resource by Hastings Technology Metals Limited and a 20.06Mt @ 1.03% TREO resource by Dreadnought Resources. The Lyons River Fault was previously interpreted as the southwestern extent of the GCCC (Slezak and Spandler 2020). The discovery of multiple carbonatite intrusions southwest of the fault by Dreadnought Resources has disproven this. Instead, ferrocarbonatite dykes were emplaced on both sides of the Lyons River Fault and large carbonatite intrusions, which are possibly the source intrusions for the GCCC, were emplaced along 17km strike of the fault itself.

Work Done: During 2024, Dreadnought drilled 28 RC holes (CBRC174-201) for a total of 3,303m at the GCC, co-funded under the 2023 Round 28 EIS grant.

Results: The EIS proposal aimed to test, primarily, the south-western portion of the Gifford Creek Carbonatite that was identified using Airborne Gravity Gradiometer (Falcon AGG), Ground Gravity, and SkyTEM surveys. This area in the carbonatites (which lacked the magnetic-ring signature of the previously drilled and identified carbonatites) had a bulk-rock geochemistry with a greater portion of CaO, with lesser MgO and Fe_2O_3, than previously drilled. These carbonatites drilled in this program were therefore more closely related to a calciocarbonatite end-member, which are often associated with Nb mineralisation at the GCC.

Whilst it was hypothesized that weaker gravity responses within the carbonatite body may indicate an area of deeper weathering, drilling suggested that it was also related to lithological variations. Several holes also intersected

a low Nb, REE and Sr unit which may be more closely related to the dolomitic marbles within the Edmund Group sediments.

Drilling returned significant mineralisation with intercepts including:

CBRC174: 38m @ 0.5% Nb_2O_5 from 58m, incl. 6m @ 1.2% Nb_2O_5 from 60m
CBRC175: 66m @ 0.4% Nb_2O_5 from 52m, incl. 15m @ 15% P_2O_5 from 103m
CBRC176: 50m @ 0.9% Nb_2O_5 from 49m, incl. 20m @ 1.3% Nb_2O_5 from 56m
CBRC179: 9m @ 0.3% Nb_2O_5 from 60m
CBRC185: 6m @ 0.2% Nb_2O_5 from 69m
CBRC189: 24m @ 0.6% Nb_2O_5 from 39m, incl. 6m @ 1.0% Nb_2O_5 from 48m
CBRC193: 57m @ 0.9% Nb_2O_5 from 51m, incl. 10m @ 1.2% Nb_2O_5 from 72m
CBRC194: 122m @ 0.7% Nb_2O_5 from 71m, incl. 26m @ 1.1% Nb_2O_5 from 99m
CBRC195: 130m @ 0.7% Nb_2O_5 from 71m
incl. 39m @ 1.3% Nb_2O_5 from 84m
incl. 4m @ 2.0% Nb_2O_5 from 86m
CBRC196: 24m @ 0.7% Nb_2O_5 from 81m, incl. 8m @ 1.3% Nb_2O_5 from 81m
CBRC197: 28m @ 0.8% Nb_2O_5 from 66m, incl. 14m @ 1.0% Nb_2O_5 from 71m
CBRC200: 95m @ 0.9% Nb_2O_5 from 48m
incl. 8m @ 1.7% Nb_2O_5 from 72m
incl. 20m @ 1.4% Nb_2O_5 from 102m
CBRC201: 98m @ 0.7% Nb_2O_5 from 54m, incl. 41m @ 1.1% Nb_2O_5 from 85m

Additional high-grade niobium mineralisation was successfully intersected at both Stinger and Rocky Road, extending their footprints to ~2,000m x 350m (Stinger) and ~1,000m x 200m (Rocky Road) respectively. Fresh niobium mineralisation is interpreted to be potentially hosted in a dyke-like geometry with a southwest dip within the Gifford Creek Carbonatite, similar to Catalao II Complex in Brazil.

Conclusion:

The drill program successfully fulfilled the purpose of its design by:

1. Providing evidence for the carbonatite intrusion extending beyond the original interpreted extent.
2. Testing geophysical targets identified through gravity and SkyTEM surveys.
3. Successfully extending the high-grade niobium mineralisation within the Stinger and Rocky Road zones.

Overall, the program was successful with results warranting follow up drilling. Next steps may include wide spaced fenceline drilling, as well as deep diamond drill holes across the Stinger zone.

To date, all drilling undertaken by Dreadnought has targeted the weathered zone, as the internal model for REE and Nb mineralisation involved supergene enrichment as a key factor to form high-grade mineralisation. However, drilling has also intersected niobium mineralisation within fresh carbonatite. Therefore, confirming the presence of fresh Nb-enriched dykes and testing their extents would be of great economic benefit as enriched fresh rock may mean the mineralisation is open at depth.

An EIS co-funded drilling grant for up to $180,000 has already been secured by Dreadnought in October 2024, which will go towards two deep diamond drill holes to test the Stinger Niobium Discovery.

2. Introduction

The Gifford Creek Carbonatite (GCC) and the Yin Ironstones together form one of the largest alkali-carbonatite complexes in the world. Carbonatite intrusions are known globally to host several different commodities including rare earths, niobium, phosphate, titanium and scandium, often as separate deposits within the same intrusion. Examples of this include Mt Weld in Australia, Ngualla in Tanzania, Araxa in Brazil, and Bayan Obo in China.

Since 2021, Dreadnought Resources has confirmed through RC and diamond drilling multiple REE-Nb discoveries, including "Yin", a major dyke-hosted rare earth element (REE) deposit in the Gifford Creek Carbonatite Complex (GCCC) of the Gascoyne Province, and "C3", a supergene weathered carbonatite plug containing an inferred maiden resource of 10.84Mt @ 1.00% TREO. Three zones of mineralisation have been confirmed at the GCC (C3, Stinger and Rocky Road) with significant niobium intercepts in the Stinger Zone.

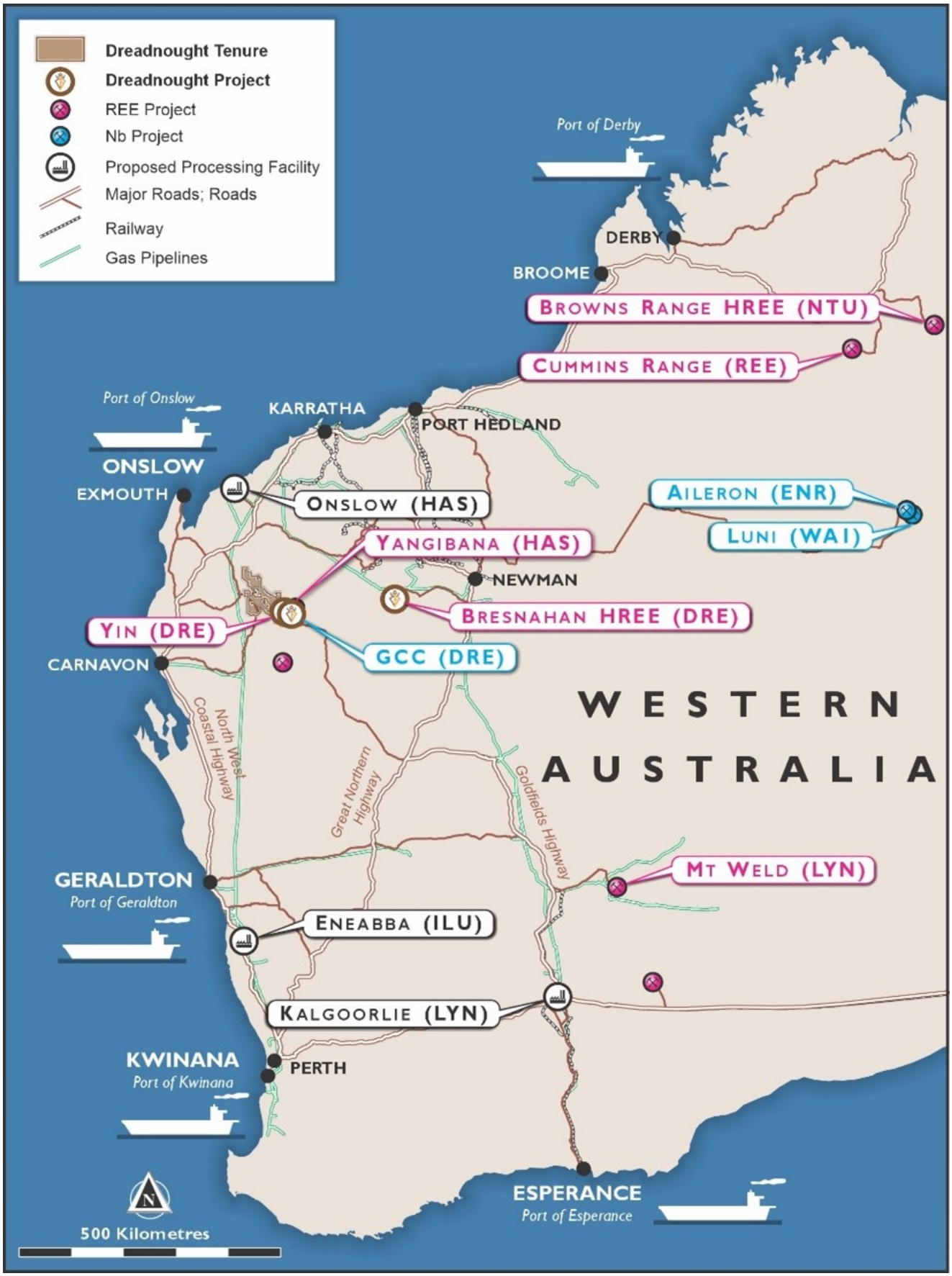
Recent drilling was completed at the GCC supported by a co-funded drilling grant of up to $180,000 under the Geological Survey of Western Australia's Exploration Incentive Scheme (EIS) (Round 28, 2023). The primary aim of the EIS-supported drilling is to understand the economic potential of the central carbonatite intrusions in the Gifford Creek Carbonatite Complex. This report outlines the results of this drilling program, which involved the completion of 28 reverse circulation (RC) drillholes from July to September 2024.

3. Location and Access Details

The GCC is part of the wider Mangaroon Project, a collection of 30 tenements spanning ~4800km^2. The Mangaroon Project is situated approximately 250 kilometers to the southeast of Exmouth, at the confluence of the Upper Gascoyne, Carnarvon, and Ashburton shires, within Western Australia's Gascoyne Region.

Access to the project area can be obtained through several routes: from Carnarvon or Gascoyne Junction via the Ullawarra/Gifford Creek Road, from Exmouth via the Towera Road branching off the Great Northern Highway, or from Meekatharra via Mount Augustus along national route 95. The terrain consists of gently undulating landscapes featuring incised drainage patterns and multiple creek crossings accessible by four-wheel drive vehicles.

The GCC, where RC drilling was completed, is located ~3km to the NW of, and parallel to the Cobra-Gifford Creek Road. The GCC extends 17km southeast of the entrance to the Yangibana Mine ('Borrow Pit'), ending just north of the Gifford Creek homestead.



Dreadnought Resources' Mangaroon Project Location Map with REE Projects in Western Australia

4. Tenement Details

Tenement Information

Tenement	Grant Date	Expiry Date	Holder	Expenditure ($)	Area Size (KM2)	Area Size (BLK)
E 09/2448	26/10/2021	25/10/2026	DREADNOUGHT EXPLORATION PTY LTD	300000	560	200
E 09/2450	24/01/2022	23/01/2027	DREADNOUGHT EXPLORATION PTY LTD	137334	350	125

5. Geology

5.1 Regional Geology

The Mangaroon REE Project lies in the Gascoyne region of Western Australia between the Yilgarn Craton and the Pilbara Craton. The Gascoyne geological province consists of a basement of Neoarchean to Palaeoproterozoic granitic gneisses (the Glenburgh Terrane) overlain by various Proterozoic metasedimentary and granitic rocks (Johnson et. al 2011). The Glenburgh Terrane was accreted to the Pilbara Craton during the Ophthalmian Orogeny (c. 2215–2145 Ma). The combined Pilbara Craton and Glenburgh Terrane then collided with the Yilgarn craton during the Glenburgh Orogeny (c. 1950 Ma). Subsequent regional deformation occurred during the Capricorn Orogeny (c. 1820–1770 Ma) and Mangaroon Orogeny (c. 1680–1620 Ma) (Sheppard et. al 2010, Sheppard et. al 2016). The Gascoyne Province has recorded six tectono-thermal events between 2005–570 Ma and contains numerous crustal-scale faults and shear zones.

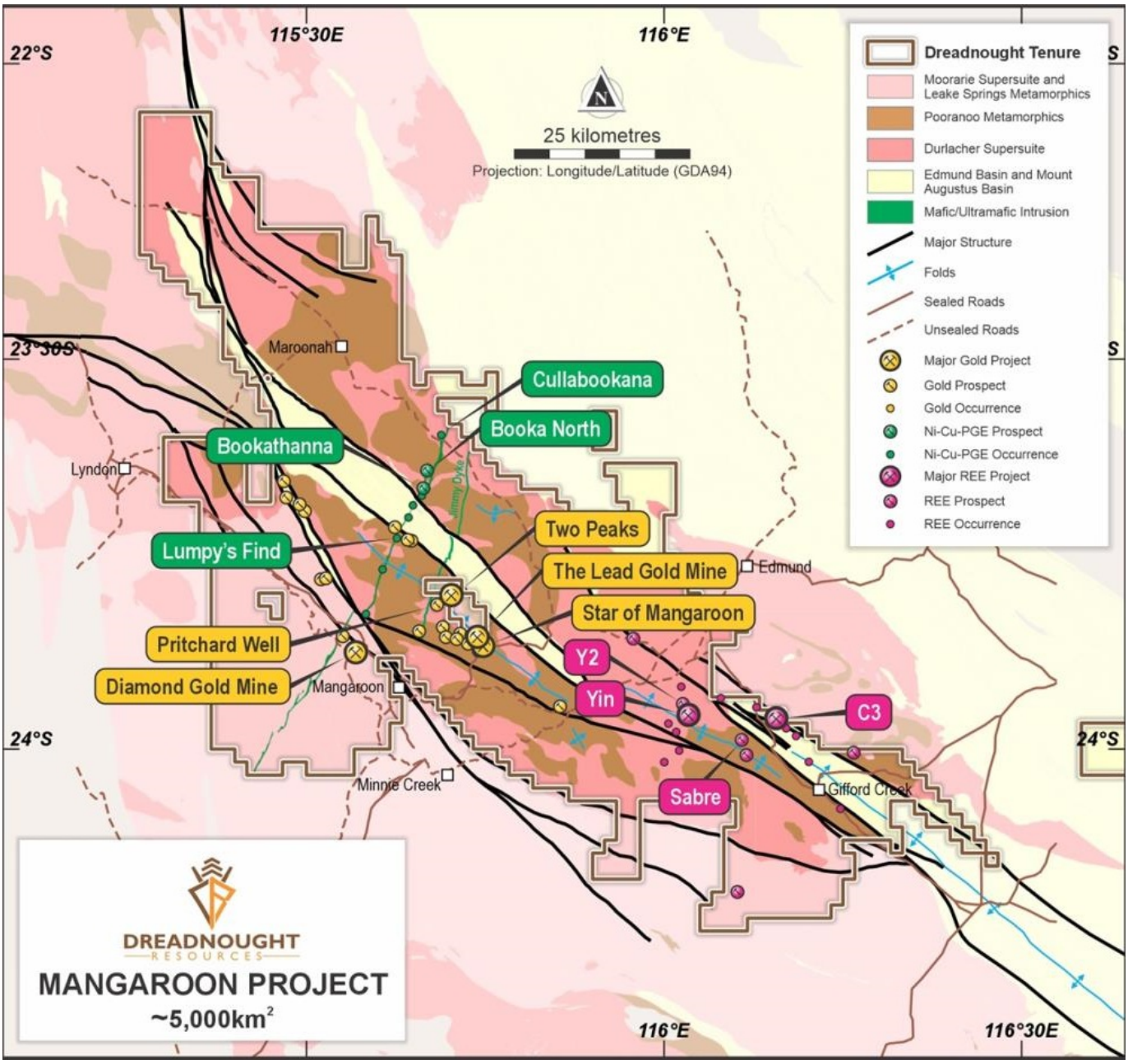
Carbonatite dykes, termed the Gifford Creek Carbonatite Complex (GCCC), were emplaced in the Gascoyne region c. 1370 Ma utilising the Lyons River Fault as a conduit through the crust (Slezak and Spandler 2020). The GCCC is a REE-prospective unit which contains the Yangibana Ironstones—REE mineralised carbonatite dykes emplaced northeast of the Lyons River Fault hosting a 29.93Mt @ 0.94% TREO resource by Hastings Technology Metals Limited and a 20.06Mt @ 1.03% TREO resource by Dreadnought Resources. The Lyons River Fault was previously interpreted as the southwestern extent of the GCCC (Slezak and Spandler 2020). The discovery of multiple carbonatite intrusions southwest of the fault by Dreadnought Resources has disproven this. Instead, ferrocarbonatite dykes were emplaced on both sides of the Lyons River Fault and large carbonatite intrusions, which are possibly the source intrusions for the GCCC, were emplaced along 17km strike of the fault itself.

5.2 Local Geology

The GCC is almost entirely covered by transported material and is interpreted as the central intrusion of the GCCC. The GCC was previously interpreted to be a series of discrete carbonatite plugs (named C1-5) due to ring-shapes features in the magnetics. However, recent geophysical advancements utilising Falcon AGG and Ground Gravity surveys have shown it to be a continuous, elongate 17x1km intrusion. The GCC contains complex mineralogy with carbonatite of different compositions, including calciocarbonatite, magnesiocarbonatite, silicocarbonatite, and ferrocarbonatite. Drilling has also intersected a potassic syenite. Whilst the majority of the GCC is unmineralised, with an average grade of 0.2-0.3% TREO, high-grade pockets exist across the length of the intrusion with elevated rare earths, niobium, phosphate, titanium and scandium. Material from the C3 target has undergone mineralogical and geochemical studies and, as such, the prospect described below in greater detail.

C3:
C3 is a deposit of 10.84Mt @ 1.00% TREO, in a 400x400m zone of the GCC. C3 is primarily composed of silicocarbonatite, with embedded anteliths of syenite, calciocarbonatite, ferrocarbonatite, magnesiocarbonatite. Anteliths, enclaves produced at different evolutionary stages from the same source are abundant throughout. Major REE-bearing phases are hosted in pseudomorphic burbankites, with several Sr-Ba-REE minerals forming from the original magmatic burbankites. Minor REE-bearing phases include fluorapatite and monazite.

Overlying the fresh carbonatite units at C3 is a 30–100 metre thick REE mineralised clay or clay-dominant saprolite weathering profile, that has stripped out primary carbonates and other minerals soluble by surficial fluids, resulting in a ~3x grade increase compared to the fresh material. This clay-dominant weathering profile generally consists of an upper magnetic red clay overlying yellow, non-magnetic clays, then a lower brown-yellow clay or clay-dominant saprolite overlying saprock or fresh rock.



Mangaroon Project Regional Geology

6. Previous Exploration

The area containing Rare Earth Elements (REE) potential in the Mangaroon Project has been primarily explored by three parties: Newmont Mining LTD, who conducted gold exploration (Marshall, 1991); Hallmark Gold N.L., who targeted the area for kimberlite-hosted diamonds; and Encounter Resources LTD, who targeted the area for calcrete-hosted uranium mineralisation.

Newmont Mining Ltd: (A5688, April 1975)

Exploration over bands of gossanous carbonate-rich ironstones within Lower Proterozoic mica schists,gneisses and granitic rocks of the Wyloo Group concluded that the ironstones contain anomalous amounts of lead,zinc and barium. Sampling indicated five positions of potential importance Two were tested without significanct result.

Newmont Mining Ltd 1989–1991

Newmont Mining explored the Gifford Creek, Lyon River, Durlacher Creek area for gold, REE, and diamonds in 1989 and 1990. In 1990–1991, Newcrest Mining, Harmon Resources, and Quicksilver Resources (Cooper, 1991) completed regional bulk geochemical exploration, with the main target being diamonds. Tenement E09/398 covered the area containing Encounter Resources Stone Tank tenement (E09/1296). A bulk sample (20 tonnes) at the Durlacher Creek/Lyons River intersection contained:

1. Anatase, barite, monazite, tourmaline, zircon, and rutile components from granite and granite-pegmatite rocks.
2. Amphibole, epidote, almandine (Fe, Al), and spessartine (Mn, Al) garnets, and ilmenite from a metamorphic (basement mica schist) association.
3. Gold and scheelite associated with columbite (a pegmatitic mineral which is an Fe, Mn, Nb, Ta oxide).
4. No microdiamonds or lamprophyric/kimberlitic indicators were found.

Newmont Mining completed gold exploration on E09/398 in 1989 and 1990 (Marshall, 1991) under the Gifford Creek project. Reconnaissance sampling at Durlacher creek included stream sediment, lag, and rock chip samples. Stream sediments were not anomalous above background; however, rock chip sampling had rare earth element (REE) anomalism and weak base metal anomalism. Ironstones after carbonatite dykes were the interpreted hosts for REE anomalism, and "felsic" dykes were considered the host of elevated Nb. Newmont suggested there were carbonatite plugs in the project area due to the presence of REE and Nb-bearing veins (see block model in Figure 7 from Elliot et al., 2018), though they remained undiscovered. No significant gold values were returned from the stream sediment sampling programs.

Newmont drilled a total of 19 RAB holes (428m) in 1990 to follow up the anomalous results from rock chip sampling and traverse magnetic and radiometric anomalies. The drill results had anomalous results of Zn (max 160 ppm), Cu (max 150 ppm), Pb (max 191 ppm), Sb (max 2.4 ppm), W (max 14 ppm), and U (max 62 ppm). The eastern side of the drilling intersected transported sediments and was unlikely to have tested the basement rocks. Petrographic descriptions confirmed sericite-biotite-quartz-feldspar±tourmaline foliated pegmatite from Hole 7 (11-12m) and Hole 8 (3-4m). No follow-up was recommended.

A second phase of RAB drilling (36 holes) was completed (1991) at the intersection of Durlacher Creek and Lyon River to test for gold anomalism. After an initial surface sample of 37ppb Au, none of the drilling returned greater than 5ppb Au. End of hole geology was petrographically described as mylonitised quartz-biotite gneiss (DUR-03) after granite and mylonitised former adamellite (DUR-12).

Hallmark Gold N.L. (A45020, Jun 1995) (A49576, Oct 1996)
Hallmark Gold N.L. acquired an area for exploration following the recognition of a suite of "kimberlite" dykes along the Lyons River. They conducted bulk stream sediment and rock sampling targeting diamond and mineral indicators and sent rock chip samples taken from the Gifford Creek Alkaline Complex to AMDEL for petrological work.

Diamond exploration was abandoned after petrology on rock chips determined that they were carbonatites and lamprophyres, not kimberlites. This did, however, reveal the location of several carbonatite intrusions along the Lyons River. Two of the carbonatite rock-chips taken by Hallmark Gold N.L. are on current Dreadnought tenements, one of which is at C4, and the other is further along the river and currently untested.

Encounter Resources Ltd (A78086, Feb 2007–Feb 2008) (A81867, Feb 2008–Jan 2009)
Encounter Resources targeted sections of the Mangaroon REE Project area for calcrete-hosted uranium mineralization. They used airborne radiometric data, surface sampling, and RC drilling to test the area. They noted schistose zones possibly after pegmatite, which they interpreted to be the source of anomalous REE geochemistry.

Ashburton Minerals Ltd (A79353, June 2007–May 2010)
Ashburton Minerals LTD conducted surface sampling and remote sensing targeting U, Au, and base metals. The surface sampling program covered the northern section of Yin, and Ashburton Minerals Ltd successfully identified the potential for U and REE mineralization in "ultramafic, carbonatite-associated dykes." Ashburton Minerals Ltd ultimately surrendered the project due to financial and time constraints.

Dreadnought Resources (2021-present)
Since 2021, Dreadnought has completed significant work within the project area. Work completed to date includes -

- Rock chipping and mapping.
- Stream sediment, lag, and soil sampling programs.
- RC drilling over the deposits and regional targets.
- Diamond drilling programs at the Yin, Y2, and Carbonatite areas.
- Infill resource drilling at Yin and the C3 carbonatite intrusion.
- Maiden resource estimation at the Yin prospect.
- Resource estimation for the C3 Carbonatite Deposit.
- Ethnographic and archaeology survey conducted by Brad Goode & Associates.
- Aeromagnetic and radiometric surveys.
- WV3 satellite imagery.
- Airborne geophysical survey over the Yin and Carbonatite areas by SkyTEM; R72504.
- Falcon AGG (Airborne Gravity Gradiometry) survey completed by Xcalibur Aviation; R72504.

- Ground gravity survey completed by Atlas Geophysics.
- IP survey completed by GAP Geophysics Exploration Services.
- Mineralogical and petrographic work, including two honours theses.
- Metallurgical studies.

7. Current Exploration

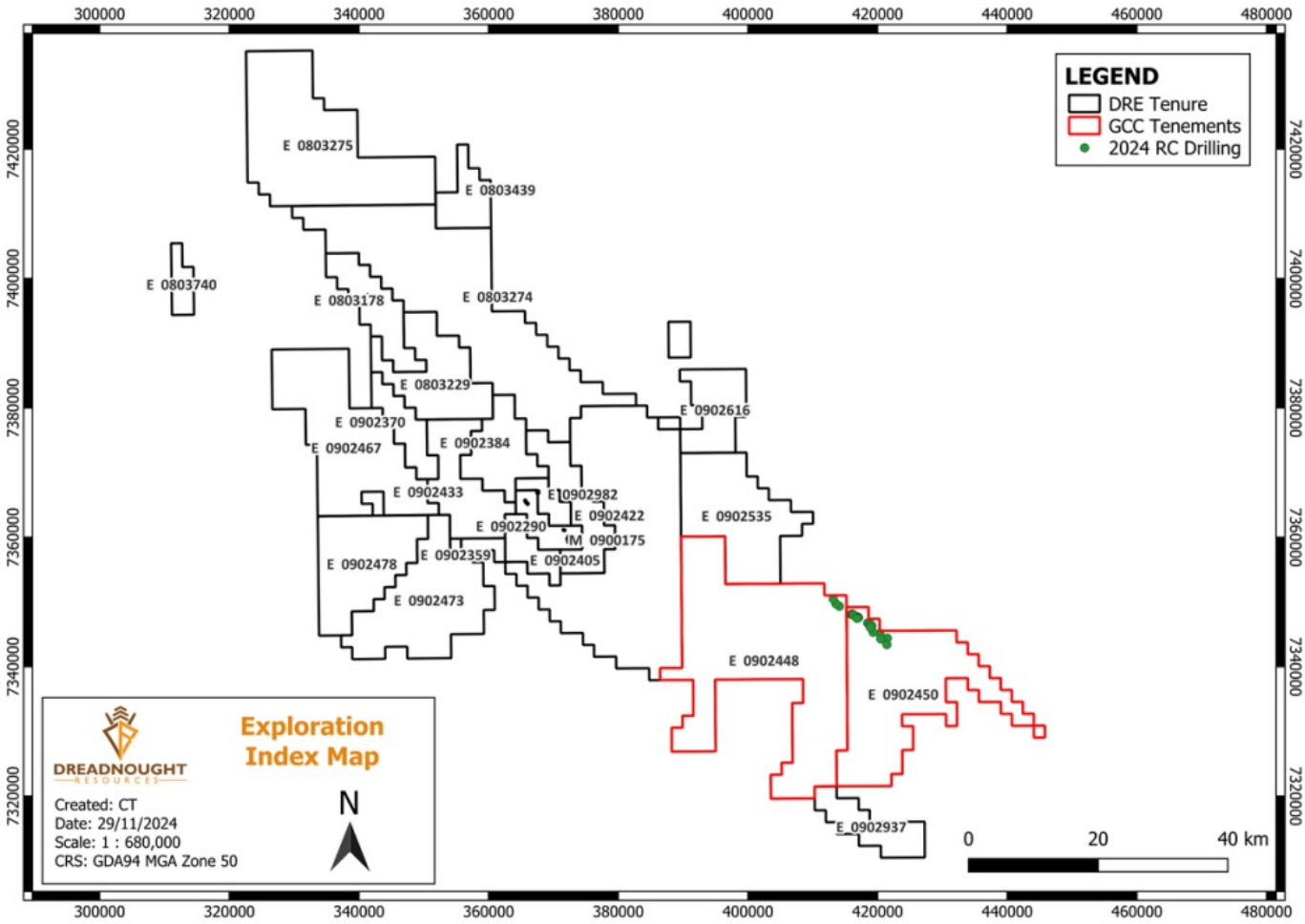
During 2024, Dreadnought drilled 28 RC holes (CBRC174-201) for a total of 3,303m at the GCC, co-funded under the EIS grant.

Current Exploration Summary

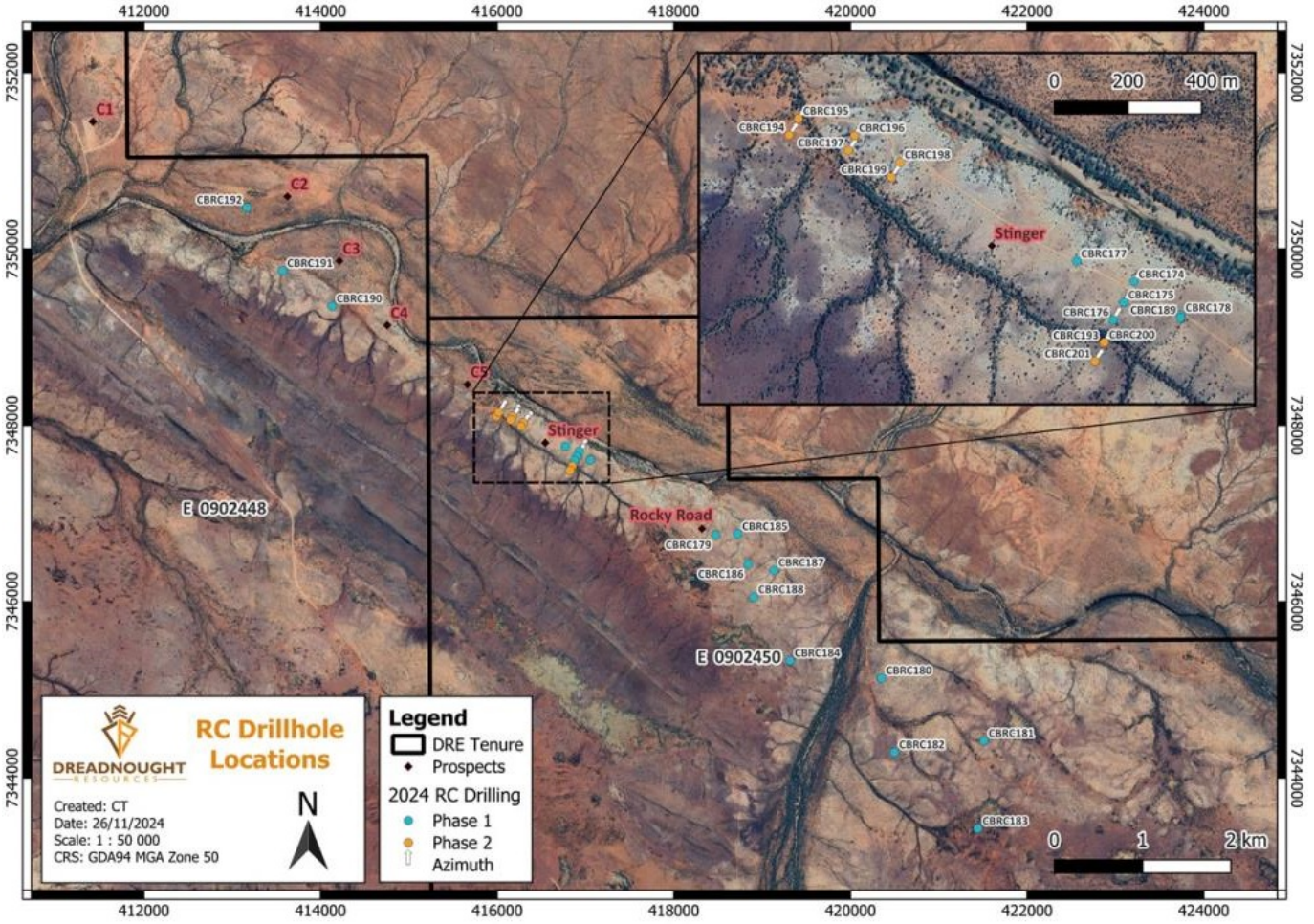
Introduction and Purpose of Drilling
Reverse circulation (RC) drilling was conducted at the Carbonatite REE-Nb prospect under the EIS. Drilling was undertaken with the following objectives:

1. Confirm whether the GCC extends to the southeast beyond the previous drilling.
2. Test regional geophysical targets believed to be zones of deeper weathering or different phases of carbonatite.
3. Test for extensions of the Stinger and Rocky Road zoned which contain tick intervals of Nb.
4. Provide further material for future mineralogical and metallurgical characterisation.

As detailed in the 'Local Geology' section, the GCC was originally interpreted to be a series of discrete carbonatite plugs. However, geophysical methods including airborne (Falcon) and ground-based gravity, as well as SkyTEM, were employed in late 2023 and have assisted in redefining the interpreted extents of the intrusion, with the GCC now interpreted to be a single continuous body. Drilling sought to confirm the new interpretation, as well as target subtle variations in geophysical response within the intrusion that may indicate variations in weathering or lithological density.
The program also sought to confirm and extend niobium enriched carbonatite at the Stinger and Rocky Road zones.



Exploration Index Map



Drillhole Location Map

8. Current Exploration Summary

8.1 Drilling

Drilling Details

Topdrill was contract to complete the drilling with 28 RC holes, for a total of 3,303m, drilled as part of the program with a truck mounted Schramm drill rig. Drilling was carried out from 19/07/2024 to 04/09/2024, with drilling completed as two phases separated by a 4-week drill break to allow for assay return. The first phase comprised 4 holes designed to confirm results from previous wide spaced (320m x 160m) first pass drilling at Stinger and an additional 15 holes designed to confirm the extents of the GCC and test regional geophysical targets believed to be zones of deeper weathering or different phases of carbonatite. The second phase consisted of a further 9 RC holes which sought to confirm fresh mineralisation and extend oxide enriched mineralisation at Stinger.

Drill holes were be logged by a geologist, recording geological information including but not limited to; lithology, weathering, grain size, texture, alteration and mineralisation. Logging was assisted by pXRF data, with readings taken every 1-2 metres. A data review was conducted on the receipt of lab assays which reveal a strong correlation between pXRF and assay results, with a slight upgrade in grade observed from the assays.

Drilling details are shown in Table 1.

Table1. Drilling details of drillholes.

Hole ID	Hole Type	EOH Depth (m)	Drill Date
CBRC174	RC	96	19/07/2024 - 20/07/2024
CBRC175	RC	126	20/07/2024
CBRC176	RC	108	21/07/2024
CBRC177	RC	72	21/07/2024
CBRC178	RC	55	21/07/2024 - 22/07/2024
CBRC179	RC	120	22/07/2024
CBRC180	RC	108	23/07/2024 - 24/07/2024
CBRC181	RC	96	24/07/2024
CBRC182	RC	108	24/07/2024 - 25/07/2024
CBRC183	RC	90	25/07/2024
CBRC184	RC	90	25/07/2024
CBRC185	RC	102	26/07/2024
CBRC186	RC	66	26/07/2024
CBRC187	RC	90	27/07/2024
CBRC188	RC	114	27/07/2024
CBRC189	RC	108	28/07/2024 - 29/07/2024
CBRC190	RC	72	29/07/2024
CBRC191	RC	72	29/07/2024
CBRC192	RC	102	30/07/2024
CBRC193	RC	108	27/08/2024 - 28/08/2024
CBRC194	RC	186	29/08/2024
CBRC195	RC	210	30/08/2024
CBRC196	RC	168	30/08/2024 - 31/08/2024
CBRC197	RC	168	31/08/2024
CBRC198	RC	168	01/09/2024 - 02/09/2024
CBRC199	RC	162	02/09/2024 - 03/09/2024
CBRC200	RC	186	03/09/2024
CBRC201	RC	152	04/09/2024
Total	3,303		

Collar details are shown in Table 2.

Table 2. Collar details of drillholes.

Hole ID	Easting	Northing	RL	Dip	Azimuth
CBRC174	416930	7347706	310	-60	31
CBRC175	416902	7347650	309	-60	32
CBRC176	416874	7347602	308	-60	29
CBRC177	416774	7347762	308	-90	0
CBRC178	417058	7347614	310	-90	0
CBRC179	418475	7346758	312	-90	0
CBRC180	420351	7345134	312	-90	0
CBRC181	421512	7344424	317	-90	0
CBRC182	420498	7344292	315	-90	0
CBRC183	421442	7343429	317	-90	0
CBRC184	419316	7345334	313	-90	0
CBRC185	418723	7346770	309	-90	0
CBRC186	418842	7346427	313	-90	0
CBRC187	419133	7346357	309	-90	0
CBRC188	418903	7346052	314	-90	0
CBRC189	417057	7347608	310	-90	0
CBRC190	414130	7349350	305	-90	0
CBRC191	413569	7349755	301	-90	0
CBRC192	413164	7350471	302	-90	0
CBRC193	416847	7347539	306	-60	32
CBRC194	415993	7348105	303	-61	33
CBRC195	416019	7348150	303	-60	31
CBRC196	416171	7348104	305	-61	32
CBRC197	416154	7348063	303	-61	33
CBRC198	416295	7348031	305	-60	30
CBRC199	416271	7347990	303	-60	30
CBRC200	416850	7347541	306	-60	30
CBRC201	416824	7347489	308	-60	30

NOTE: Collar locations measured using a DGPS and are reported in GDA94 MGA50. Dip and azimuth measurements detailed were recorded at surface by Topdrill using an Axis Champ North Seeking Gyro. An azimuth of 0 is recorded for all vertical holes.

Drilling Results

Drilling intersected a thin layer of calcrete or colluvium, paleochannel to 30-50m, then weathered carbonatite to around 100m, transitioning to fresh carbonatite at the bottom of hole. The paleochannel was ubiquitous due to the proximity to the Lyons River, and the weathered zone above fresh carbonatite extended to depths up to 166m. This significant zone of weathering was considerably deeper than expected (when compared to previous drill programs in this carbonatite) and extremely encouraging as a high portion of the economic mineralisation in carbonatites are in the weathered zone. Widths of up to 130m of continuous mineralised material were returned from the assays. Follow up drilling was conducted on encouraging Niobium mineralisation.

The EIS proposal aimed to test, primarily, the south-eastern portion of the Gifford Creek Carbonatite that was identified using Airborne Gravity Gradiometer (Falcon AGG), Ground Gravity, and SkyTEM surveys. This area in the carbonatites (which lacked the magnetic-ring signature of the previously drilled and identified carbonatites) had a bulk-rock geochemistry with a greater portion of CaO, with lesser MgO and Fe_2O_3, than previously drilled. These carbonatites drilled in this program were therefore more closely related to a calciocarbonatite end-member, which are often associated with Nb mineralisation at the GCC. Several holes also intersected a low Nb, REE and Sr unit which may be more closely related to the dolomitic marbles within the Edmund Group sediments.

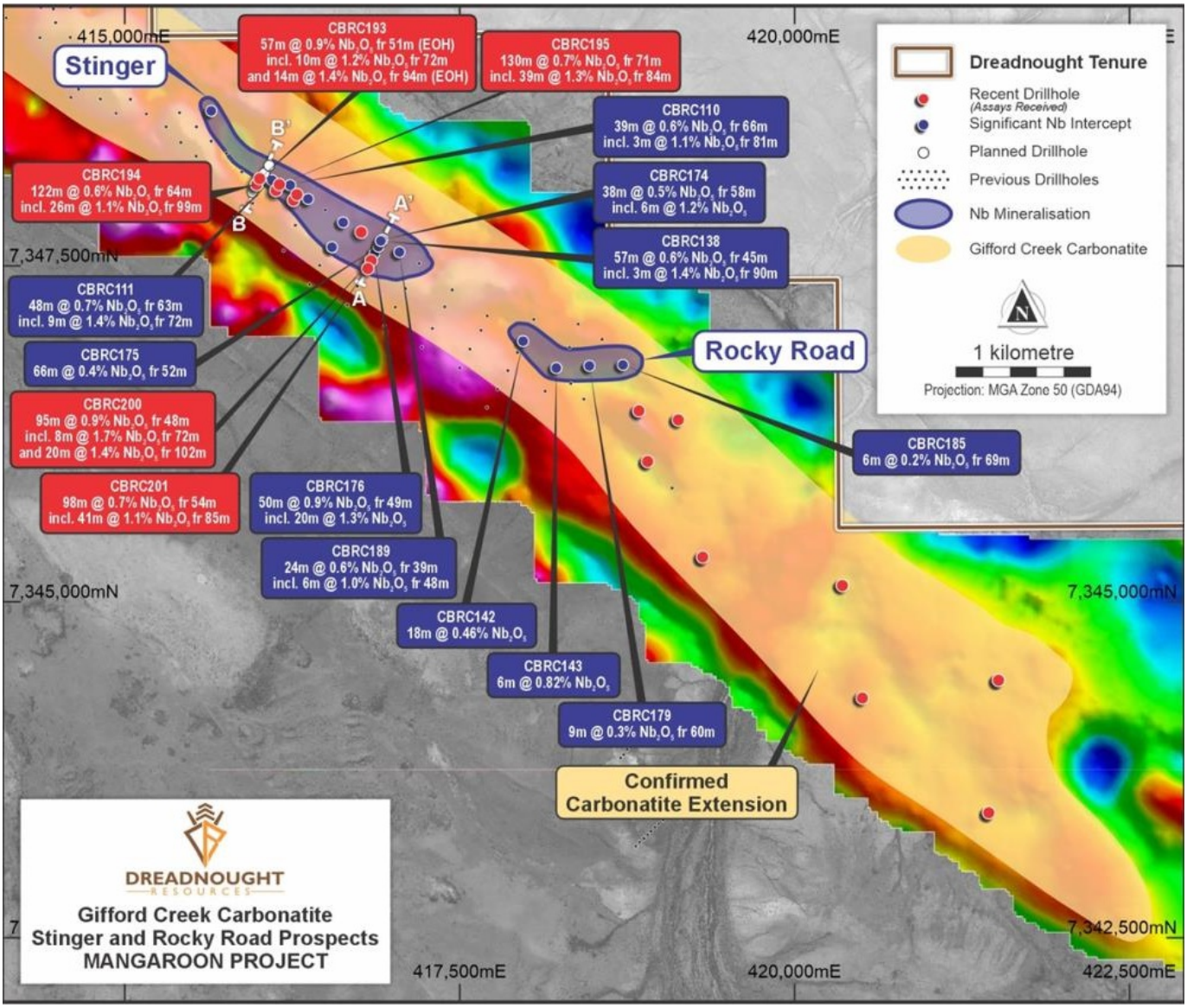
Whilst it was hypothesized that weaker gravity responses within the carbonatite body may indicate an area of deeper weathering, drilling suggested that it was also related to lithological variations.

Additional high-grade niobium mineralisation was successfully intersected at both Stinger and Rocky Road, extending their footprints to ~2,000m x 350m (Stinger) and ~1,000m x 200m (Rocky Road) respectively. Fresh niobium mineralisation is interpreted to be potentially hosted in a dyke-like geometry with a southwest dip within the Gifford Creek Carbonatite, similar to Catalao II Complex in Brazil.

Drilling returned significant mineralisation with intercepts including:

CBRC174: 38m @ 0.5% Nb_2O_5 from 58m, incl. 6m @ 1.2% Nb_2O_5 from 60m
CBRC175: 66m @ 0.4% Nb_2O_5 from 52m, incl. 15m @ 15% P_2O_5 from 103m
CBRC176: 50m @ 0.9% Nb_2O_5 from 49m, incl. 20m @ 1.3% Nb_2O_5 from 56m
CBRC179: 9m @ 0.3% Nb_2O_5 from 60m
CBRC185: 6m @ 0.2% Nb_2O_5 from 69m
CBRC189: 24m @ 0.6% Nb_2O_5 from 39m, incl. 6m @ 1.0% Nb_2O_5 from 48m
CBRC193: 57m @ 0.9% Nb_2O_5 from 51m, incl. 10m @ 1.2% Nb_2O_5 from 72m
CBRC194: 122m @ 0.7% Nb_2O_5 from 71m, incl. 26m @ 1.1% Nb_2O_5 from 99m
CBRC195: 130m @ 0.7% Nb_2O_5 from 71m
incl. 39m @ 1.3% Nb_2O_5 from 84m
incl. 4m @ 2.0% Nb_2O_5 from 86m

CBRC196: 24m @ 0.7% Nb_2O_5 from 81m, incl. 8m @ 1.3% Nb_2O_5 from 81m
CBRC197: 28m @ 0.8% Nb_2O_5 from 66m, incl. 14m @ 1.0% Nb_2O_5 from 71m
CBRC200: 95m @ 0.9% Nb_2O_5 from 48m
incl. 8m @ 1.7% Nb_2O_5 from 72m
incl. 20m @ 1.4% Nb_2O_5 from 102m
CBRC201: 98m @ 0.7% Nb_2O_5 from 54m, incl. 41m @ 1.1% Nb_2O_5 from 85m



Drillholes Map over Gravity

8.2 Mineralogy & Geochronology

Ten RC samples from the recent drilling have been submitted for XRD analysis. Samples were selected from a range of holes across the Stinger prospect and encompass a mix of oxide, saprock and fresh, different lithologies and Nb_2O_5 grades. Results from this study will aid the characterisation of the Stinger prospect, the mineralogical variability within it and its genetic relationship within the broader GCC. Additionally, XRD will help identify the niobium species that host the mineralisation, which will influence both economic and metallurgical prospectivity. XRD results are still currently pending and will be submitted upon receipt.

Numerous intervals returned high zirconium concentrations. These samples will be collected and sent for geochronology via zircon dating with the aim to provide an age for the GCC. This work is planned for 2025 as part of the ARC Linkage-MRIWA carbonatite project.

9. Conclusion and Recommendations

The drill program successfully fulfilled the purpose of its design by:

1. Providing evidence for the carbonatite intrusion extending beyond the original interpreted extent.
2. Testing geophysical targets identified through gravity and SkyTEM surveys.
3. Successfully extending the high-grade niobium mineralisation within the Stinger and Rocky Road zones.

Overall, the program was successful with results warranting follow up drilling. Next steps may include wide spaced fenceline drilling, as well as deep diamond drill holes across the Stinger zone. To date, all drilling undertaken by Dreadnought has targeted the weathered zone, as the internal model for REE and Nb mineralisation involved supergene enrichment as a key factor to form high-grade mineralisation. However, drilling has also intersected niobium mineralisation within fresh carbonatite. Therefore, confirming the presence of fresh Nb-enriched dykes and testing their extents would be of great economic benefit as enriched fresh rock may mean the mineralisation is open at depth.

An EIS co-funded drilling grant for up to $180,000 has already been secured by Dreadnought in October 2024, which will go towards two deep diamond drill holes to test the Stinger Niobium Discovery.

10. References

Cooper, R. W. (1991). Edmund Diamond Joint Venture Annual Report 1991, WA. Newcrest Mining Limited. Report number A36887.

Elliott, H. A. L., Wall, F., Chakhmouradian, A. R., Siegfried, P. R., Dahlgren, S., Weatherley, S., Finch, A. A., Marks, M. A. W., Dowman, E., Deady, E. (2018). Fenites associated with carbonatite complexes: A review. Ore Geology Reviews, 93, 38–59. https://doi.org/10.1016/j.oregeorev.2017.12.003

Johnson, S. P., Sheppard, S., Rasmussen, B., Wingate, M. T. D., Kirkland, C. L., Muhling, J. R., Fletcher, I. R., Belousova, E. A. (2011). Two collisions, two sutures: punctuated pre 1950 Ma assembly of the West Australian Craton during the Ophthalmian and Glenburgh Orogenies. Precambrian Research, 3–4, 239–262.

Marshall, J. (1991). GIFFORD CREEK E09/396 - 398, E09/413 - 415, E09/442, E09/443. Annual report for the period to 31/12/90 volume one. Newmont Australia Limited. Report number A32886.

Sheppard, S., Bodorkos, S., Johnson, S., Wingate, M., Kirkland, C. (2010). The Paleoproterozoic Capricorn Orogeny: intracontinental reworking not continent-continent collision (108). Geological Survey of Western Australia.

Sheppard, S., Fletcher, I. R., Rasmussen, B., Zi, J.-W., Muhling, J. R., Occhipinti, S. A., Wingate, M. T. D., Johnson, S. P. (2016). A new Paleoproterozoic tectonic history of the eastern Capricorn Orogen, Western Australia, revealed by U–Pb zircon dating of micro-tuffs. Precambrian Research, 286, 1–19. https://doi.org/10.1016/j.precamres.2016.09.026

Slezak, P., & Spandler, C. (2020). Petrogenesis of the Gifford Creek Carbonatite Complex, Western Australia. Contributions to Mineralogy and Petrology, 175, 28. https://doi.org/10.1007/s00410-020-1666-3

11. Appendices

No Appendices as text are available